Eyes Bring out Effectiveness? Applying Eye-Tracking to Explore Consumers' Visual Perception of Horizontal and Vertical Menu Formats in E-commerce Website Design

1 INTRODUCTION

In recent years, scholars have initiated inquiries into website menus (Naylor, 2016; Ouyang & Zhou, 2018), proposing that the fundamental design of menus should underscore both breadth and depth (Naylor, 2016). With technological progress, menus have undergone alterations in diverse presentation formats. In a prior investigation into medical website design, it was determined that the quantity of screens necessary to navigate to the desired content exerted the most significant influence on usability (Rotondi et al., 2021). Every website necessitates a rapid and effective navigation system(Leuthold et al., 2011), as it is crucial for ensuring that users can easily and efficiently find the information they are looking for, thereby enhancing their overall experience and satisfaction.. Many books on website or user interface design broadly categorize navigation systems into two consecutive conceptual tasks (Deaton, 2003; Garrett, 2022; Kalbach, 2007; Lynch et al., 1999; Rosenfeld & Morville, 2002; Shneiderman & Plaisant, 2010): (1) the task of organizing items into meaningful clusters and (2) the task of tour design, which involves determining how to present the problem of linking with other tour mechanisms to encourage users to navigate as efficiently and effectively as possible. Hence, the efficacy of the navigation system plays an exceedingly important role on the website. Based on Just and Carpenter's eye-mind hypothesis (Just & Carpenter, 1976), this study investigates perception and cognition, dividing them into distinct components. Eye attention is a crucial design principle for attracting people's focus(Cuddihy & Spyridakis, 2012). In this study, Eye-tracking gaze patterns provide primary data on users' visual movement and efficiently, while real-time user feedback during operation offers secondary insights into immediate user cognition and overall effectively. A review of relevant literature reveals that previous menu studies have predominantly focused on user interface design and user search behavior across different task types, with limited in-depth exploration into perception and cognition of individual menu systems. Therefore, the core of this study is to propose recommendations based on eye-tracking dynamics for user perception of mega menus.

A well-designed navigation system not only resolves the issue of getting lost promptly but also enhances information-seeking efficiency. While the Internet continues to progress rapidly (Zheng, 2015), there remains a significant number of websites employing ladder menus, also known as drop-down menus (Naylor, 2016). Consequently, recent research in navigation systems has primarily focused on delving deeper into drop-down menus. Over the years, a diverse range of menu types has been investigated (Hornof & Halverson, 2003; Kiger, 1984; Kreigh et al., 1990; Landauer, 1987; Naylor, 2016; Zaphiris, 2000). However, the relatively new mega menu has received less attention in research endeavors. Several studies in the existing literature have investigated menu design in connection with mega menus. For instance, in Hornof and Halverson's study, it was shown that the label menu and the mega menu are similar in terms of presentation formats. Their findings demonstrated that the labeled layout resulted in faster search speeds compared to the unlabeled layout. However, it is worth noting that the focus of Hornof and Halverson's study was primarily on search strategies and categorization/tagging rather than the intricacies of menu design itself (Hornof & Halverson, 2003). In 2011, Leuthold

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conducted a study on task complexity in relation to menu design in navigation systems. The study revealed that vertical menus promote quicker search behavior within a reduced timeframe. When juxtaposed with dynamic menus, vertical menus demonstrated enhanced perceptual and cognitive efficiency (Leuthold et al., 2011). Nevertheless, the study solely delved into task complexity concerning simple, vertical, and dynamic menus, providing limited insight into users' physiological perceptions. The current utilized predominant menu type is the drop-down menu, typically arranged in a vertical layout (Naylor, 2016). In recent times, the mega menu has gained traction, being implemented both vertically and horizontally. Previous studies have demonstrated that vertical menus can effectively decrease user operation time and enhance efficiency(Leuthold et al., 2011). However, research in this domain remains limited, with minimal exploration into the physiological perceptions of users regarding both horizontal and vertical menu structures. In this study, mega menus are defined as menu systems that can display lower-level links within the same screen area, providing a clear hierarchical view of the menu structure. The research further explores the design and effects of both main menus and local menus within this context.

In the domain of human-computer interaction (HCI), substantial research and scrutiny into the aesthetics of user interfaces gained momentum following the seminal study by Kurosu and Kashimura in 1995(Kurosu & Kashimura, 1995; Tuch et al., 2012). Particularly within the realm of user experience (UX) research, the aesthetics of user interfaces has emerged as a prominently discussed subject(Bargas-Avila & Hornbæk, 2011; Tuch et al., 2012). Despite numerous studies highlighting a significant correlation between usability and aesthetics(Hartmann et al., 2008; Lavie & Tractinsky, 2004; Quinn & Tran, 2010; Tractinsky et al., 2000), historical research findings still exist suggesting that aesthetics does not significantly impact usability (Hassenzahl, 2004; Van Schaik & Ling, 2009). As a result, recent studies have indicated that the presence of usability elements could potentially influence users' perceptions of a website's aesthetics. Furthermore, this notion may extend to the concept that functionality and aesthetic appeal are closely intertwined, suggesting that effective functionality is inherently aesthetically pleasing. Research on the relationship between usability and aesthetics has recently gained traction in empirical studies. Ben-Bassat et al. (2006) and Lee and Koubek (2010) have highlighted the significant impact of interface usability on users' perceptions of aesthetics. Their findings suggest that interfaces that are easier to use tend to be perceived as more aesthetically pleasing (Ben-Bassat et al., 2006; Lee & Koubek, 2010). While previous research has hinted at the notion that "beautiful things are easier to use," this idea is only partially supported by empirical evidence. More recent scholars have started to posit that usability might exert a more direct influence on user interface evaluation than aesthetics do, shaping users' overall perceptions of the interface. Therefore, while UI concentrates on the overall presentation of the interface and holds significant importance, UX encompasses broader aspects such as system accessibility, efficiency, and user enjoyment

The advancement of Internet technology has hastened the increasingly shorter development cycles of the Web(Almeida, 2017). Furthermore, the rapid evolution of the Web has expedited the formation of people's behavioral patterns online, particularly concerning the dissemination, sharing, and acquisition of information (Chang et al., 2022). In information environments, IA plays a critical role in shaping the structural organization, navigation system, and categorization labels (De Marsico & Levialdi, 2004; Peter Morville & Rosenfeld, 2006; Powell et al., 1998; Rosenfeld et al., 2015; Toms, 2002; U.S. General Services Administration; Wurman & Richard Saul, 1996). The term "information architecture" was initially defined in 1997 as "a simple set of auxiliary tools to resourcefully integrate users' needs with information"(Davenport & Prusak, 1997). Abstractly conceptualized, IA provides a structured information framework or map that enables users to navigate their individual paths to knowledge(Juvina, 2006; Wurman & Richard Saul, 1996). The menus within an Information Architecture (IA) navigation system consist of links to various pages, and the arrangement of these linked pages is commonly termed as a mini-IA, significantly influencing usability(Naylor, 2016). While menus represent just a

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portion of the overall IA, they serve as a condensed representation of the IA, encapsulating the primary IAs of the website that users can access via the menus. Therefore, this study aims to synthesize insights gleaned from prior literature, delineating them as follows:

(1) Menus significantly contribute to the usability of a website.

(2) Previous studies predominantly focused on drop-down menus as a menu format.

(3) Adopting an effective menu format can enhance users' efficiency in navigating the website.

(4) Vertical menus are particularly conducive to perceptual exploration.

In 2011, Leuthold, Schmutz, and other researchers utilized eye-tracking technology to investigate disparities in visual perception associated with employing simple menus, dynamic menus, and vertical menus. The study outcomes indicated that users required reduced attention when interacting with vertical menus, leading to faster and more successful overall navigation. Furthermore, the findings suggested that vertical menus are better suited for exploring user perception and cognition(Leuthold et al., 2011). Presently, the scope of menu studies utilizing eye-tracking devices lacks diversity, necessitating exploration of additional menu types for related research. As asserted by Jacob and Karn (2003) and supported by Nielsen and Pernice (2010), eye-tracking represents a potent method for delving into navigation design intricacies, facilitating comprehension of how users' eye movements and saccade patterns are deployed for information acquisition. Previous investigations have demonstrated that incorporating known-item seeking, while maintaining task time within manageable bounds for experimental efficacy, affords researchers deeper insights into individuals' behaviors throughout the information retrieval process (Jacob & Karn, 2003; Nielsen & Pernice, 2010). Hence, drawing upon prior research and literature, this study discerned that amalgamating both horizontal and vertical configurations within the mega menu structure could notably amplify users' effectiveness in Known-item seeking information retrieval. Historically, eye-tracking studies have often depicted users' overall scanning patterns resembling the F-shaped (or E-, Z-shaped) when navigating web pages(Nielsen, 2006; Nielsen Norman Group, 2009; Shrestha & Owens, 2009). This investigation seeks to elucidate whether aligning menu formats (i.e., vertical versus horizontal) with human oculo-physiological scanning tendencies constitutes a contributory factor in augmenting user efficiency. The F-shaped layout indicates that the primary menu is displayed horizontally, while the secondary menu is displayed vertically. Diverging from traditional horizontal or vertical layouts, this study aims to integrate the F-shaped concept into the visual breakpoints within the user experience flow of menus. By introducing breakpoints, the overall efficiency of user interactions is thereby enhanced.

In the relevant domain, it was found that in 1988, scholars conducted a study involving six types of menu layouts, ultimately demonstrating that a two-level menu structure could achieve optimal operational performance (Parkinson et al., 1988). This optimal menu prototype bears resemblance to today's mega menu layout. However, in 2016, Naylor posited that the type of menu does not affect search performance, a finding inconsistent with previous research (Naylor, 2016). Nevertheless, upon reviewing past studies, there has been only preliminary discussion on mega menus, with most experimental research relying on the duration of time to assess efficiency. Direct validation and discussion on the efficiency of menus at higher levels remain scarce. Therefore, there exists untapped potential and scope for expansion in this field.

2 LITERATURE REVIEW

2.1 Information Architecture

Information Architecture (IA) was first proposed by Richard Saul Wurman, who is known as the master of information theory (Wurman & Richard Saul, 1996). In the book "*Information Architecture for the World Wide Web"*, it is mentioned that 'we cannot ask users to take the time to learn to understand our information architecture, so we should do our best to

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build and design an information architecture that is accessible to users' (Peter Morville & Rosenfeld, 2006; Rosenfeld et al., 2015; Rosenfeld & Morville, 2002). IA is an extension of Typology(Rosenfeld & Morville, 2002; Wodtke, 2011; Wurman & Saul, 1996), which is composed of three main structures, namely context, content, and navigation. In 1997, IA was defined as a set of simple assistive tools to integrate users' needs with information resources(Davenport & Prusak, 1997). The application of IA to websites is composed of four systems, namely, organizational systems, labeling systems, navigation systems, and search systems.

In 2017, scholars mentioned that IA is an important component of a website and that IA in a website significantly affects its usability and user experience during navigation(Rosenfeld & Morville, 2002; Schmitt, 1999; Wan Mohd Isa et al., 2007). It is even suggested that IA is not just a theory, but a strategy for a website as well as an application of best practices. IA is the result of continuous iteration and negotiation between website designers and administrators (Dijck, 2003; Ruzza et al., 2017; Schmitt, 1999). In 2020, scholars emphasized the significance of information architecture within websites and recommended that it be conscientiously integrated into website navigation systems during the development phase (Tang et al., 2020). IA can be likened to a tree; with a solid foundation, the overall experience becomes more stable and reassuring. Consequently, numerous scholars have emphasized the substantial influence of a well-crafted IA on the overall user experience. In a navigation system, a menu represents a set of pages interconnected through a series of links. This interconnected structure of linked pages is commonly termed as a mini-information architecture, exerting significant influence on the usability of each touchpoint (Naylor, 2016). While the menu constitutes a subset of the overarching information architecture, it functions as a microcosm thereof, encapsulating the primary data structure of the website and facilitating rapid information retrieval for users via the menu interface.

The book categorizes users’ behaviors into two dimensions: integration and iteration and also outlines the four most common ways of finding information, including exhaustive research, exploratory seeking, refinding, and known-item seeking (Rosenfeld & Morville, 2002). Known-item seeking was initially introduced in the context of library materials seeking and is defined as the assumption that the searcher knows the information and is looking for a specific item (Wildemuth & O'Neill, 1995). In a 1989 study, the concepts of "known-item search" and "closed task" were investigated. The study highlighted that the time required for an intentional information search was shorter compared to a random or purposeless search. This phenomenon stems from the fact that focused tasks provide individuals with a clearer objective for information retrieval. Conversely, aimless searches necessitate more time as they involve sifting through a broader range of information (Marchionini, 1989). In contrast, a study in 1993 revealed that individuals engaging in Known-item seeking were more inclined to employ personalized analyses and strategies during their searches (Qiu, 1993). Synthesizing the aforementioned literature, Known-item seeking not only manages task duration, enhancing research efficiency, but also illuminates the behavioral characteristics inherent in individuals during information retrieval processes. Thus,this research aims to understand the user visual perception of known-item seeking behavior on information architecture of menu design.

Numerous theoretical concepts and definitions concerning information-seeking behavior have been examined in previous research. In 2003, scholars presented a white paper addressing four major areas critical to constructing comprehensive website information: Knowledge elicitation, Organization and structure of information, Retrieval of information, and Presentation of information (Proctor et al., 2003). Understanding users' expectations from a website during information-seeking processes can significantly enhance website satisfaction (Ping et al., 1999). Therefore, this study expands upon and delves into user-centered research to achieve a more profound understanding of website information architecture from the perspectives of both users and experts.

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2.1.1 Navigation System: Information Seeking

A navigation system plays a pivotal role in improving the usability of a website, a task inherently laden with cognitive demands for users. According to Jul (2004), a navigation system comprises four cognitive subtasks: information gathering, wayfinding, spatial knowledge acquisition, and mobility(Jul, 2004). In a web environment, when users struggle to locate desired content, they often experience a loss of orientation and become disoriented within the information space (Chang et al., 2022; Lazar et al., 2003; Levene, 2011; Nah & Davis, 2002), leading to user frustration (Lazar et al., 2003). This frustration, in turn, impacts the website's brand image, increasing the likelihood of users seeking alternative websites (Krug, 2000; Naylor, 2016; Rosenfeld & Morville, 2002) for their information needs. Hence, throughout the website development phase, strategic considerations must be given to augmenting user engagement, encouraging purchasing activities, and fostering repeat visits, thereby bolstering the website's overall integrity and commercial viability (Naylor, 2016; Nielsen, 1999). Besides underscoring the significance of user interface design (Tractinsky et al., 2000), a comprehensive approach to evaluating the website's interactivity is crucial to sustain user retention, ensuring the website's enduring functionality and efficacy over time.

Throughout the literature, diverse perspectives have been presented by scholars regarding information-seeking behavior. In 1989, Marcia Bates introduced the "Berry-picking" model, which elucidates how users formulate information needs at the outset of their search process and subsequently develop search parameters based on these needs. As the search unfolds along intricate pathways, users progressively traverse these paths to access additional information (akin to picking strawberries). Consequently, users gradually refine their understanding of their needs and modify their initial search criteria (Bates, 1989). This perspective aligns with assertions made by scholars in 2015, who propose that navigation within the information landscape is guided by individual conceptual frameworks. Users determine their subsequent actions based on task objectives and contextual elements (Jul & Furnas, 1997; Zheng, 2015). In the navigation system, users can manifest a higher degree of personal awareness in their information-seeking behavior. Users engaging in information seeking can be classified into three primary groups: Low information seekers, High information seekers, and Selective information seekers. Investigating the group of selective information seekers may reveal a greater degree of homogeneity compared to the high or low information seeker groups.

Fenichel's (1981) research examined search performance within an online database system, revealing a noteworthy correlation between users' familiarity with a website and their search efficacy (Fenichel, 1981; Kim, 2001). Moreover, users' experiences during searches significantly influence their adopted search strategies (Kim, 2001). Synthesizing these prior investigations, it becomes evident that navigation systems intricately intertwine with users' behaviors throughout information retrieval, consequently impacting their overall experience, search strategies, and efficiency. In essence, extant literature underscores the pivotal role of navigation systems within information architectures, not only facilitating users' access to information but also steering them towards optimal exploration within the digital information domain. A well-designed navigation system not only resolves the issue of getting lost promptly but also enhances information-seeking efficiency. Common navigation tools encompass menus, website maps, and navigation paths (Zheng, 2015). Despite the increased emphasis on search systems in recent years, navigation systems remain fundamental and are often deemed the "last mile" for users in accessing valuable information. Presenting website content and information structure in a straightforward format (Danielson, 2002; Tedesco et al., 2008; Zhang & Salvendy, 2001; Zheng, 2015) constitutes one of the pivotal factors contributing to a website's usability (Palmer, 2002; Zheng, 2015).

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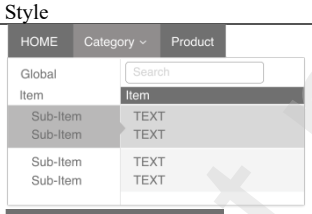
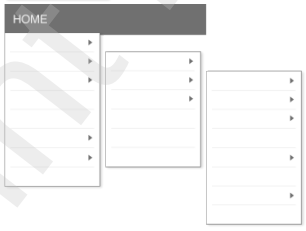
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2.1.2 *Menu Format*

Menus, as a common method of website navigation, encompass a collection of functions interlinked with various pages. The structural organization of these page links, often denoted as "information architecture," significantly influences the usability of a website (Naylor, 2016). In 2015, Jack G. Zheng assessed navigation systems using a two-dimensional framework, addressing both information structure and usability. This evaluation provided an exhaustive examination of primary navigation tools and systems then in use, thereby offering a comprehensive analysis of menus within navigation systems alongside an in-depth exploration of IA and usability. Zheng's work contributed to a more refined understanding, description, and conceptualization of menus within navigation frameworks.

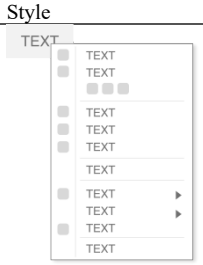
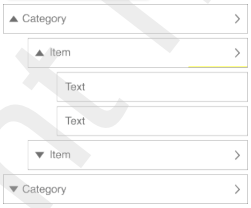
In the past, the most commonly used type of menu was referred to as a Global Menu or Global Navigation, and it served as one of the primary menu styles in navigation systems. Typically, this menu is displayed horizontally at the top of a webpage or vertically on its left side. Global menus primarily provide links to the top-level pages of a website, with sub-level options appearing as needed (Zaphiris et al., 2002; Zheng, 2015), adhering to fundamental principles of sequential menu design. However, global menus that display only select top-level links no longer align with contemporary trends. Scholars have emphasized the importance of menus containing more information-structured content. Therefore, menus incorporating multi-level structured content are termed Multi-level navigation menus or Structure Preview Designs (Zhang & Salvendy, 2001). In addition to the drop-down menu, which is the most common menu in the past few years, a variety of menu forms have begun to appear, such as fly-out menu (used for vertical menus on the left or right side), pop-up menu (at the bottom of the page), multi-level navigation bar, mega menu, fat footer, HTML sitemap page, breadcrumbs and so on. As such, this study compiles recent scholarly definitions of 'menu' and presents them in Table 1.

Table 1 Menu Style and definition.

Navigation system	Style	Definition
Global Menu (global Navigation)		It is an interface region reserved for design elements. It may include titled sections, dropdown menu, tabs, buttons, search bars and whole links. To provide a consistent means of traveling to anywhere on the website(IxDF., 2020; Nielsen & Pernice, 2010).
Drop-down menu		It provides hierarchical structure with mouse click (or mouse-over)(Andrea, 2012; Naylor, 2016; Yang & Chiu, 2022; Zheng, 2015). It is the most common form of menu(Naylor, 2016).
Fly-out menu		It is a horizontal layout of the drop-down menu, where the sub-menus 'flyout' from the side when you mouse-over (or mouse click) the item(Andrea, 2012; Yang & Chiu, 2022; Zheng, 2015). It also called a dropdown menu and used to concise the alignment of the website. This style menu is arranged to show one option at the time when the user clicks on it(Eggert & Abou-Zahra, 2015; Fatima, 2019).

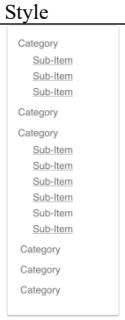
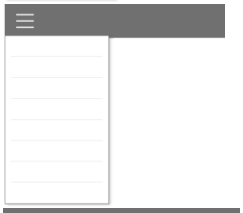
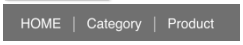
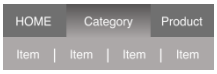
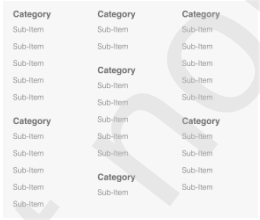
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Navigation system	Style	Definition
Pop-up menu		It is context-sensitive and is triggered at specific locations within the user interface(Pook et al., 2000). That is hidden and doesn't appear until user right-mouse click.
Mega menu (Fat menu)		It also the dropdown menus that not only contain links but also accommodate additional content(Andrea, 2012). One of its advantages is that it exposes a lower-level menu without requiring extensive navigation through nested menus(Nielsen & Li, 2017; Ouyang & Zhou, 2018).
Accordion menu		It is the vertical menus in which clicking on the main item expands the section beneath it. Clicking on the top link again collapses the section(Andrea, 2012). While longer pages can enhance user experience, accordions streamline content presentation by reducing scrolling. However, they also introduce an interaction cost by necessitating users to make decisions regarding topic headings(Loranger, 2014).
Tree menu		Stacking information content through hierarchical structures to establish a tree-like system architecture. However, an increasing number of online retailers have abandoned the use of tree menus as a means of user navigation. This is because excessive folding and collapsing can lead to users losing their sense of orientation(Kalbach, 2007; Rivers, 2020).
Hierarchical menu		It is based on hierarchy of facet attributes to creates a navigation menu. It is typically employed for categories that contain subcategories(Algolia, 2023).

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Navigation system	Style	Definition
Labelled menu		Like mega menu. Through clearly labeled layouts visual hierarchy to group the items.(Hornof & Halverson, 2003)
Hamburger Menu (Hidden Menu)		Vertical version of drop-down menu completely provides the hierarchical structure with mouse click actions(Yang & Chiu, 2022). Its icon like hamburger(FEDEWA, 2021; Khadka, 2021; Yang & Chiu, 2022). When user clicks the hamburger icon, the menu will be closed. Therefore, it also called the hidden menu.
Tab Bars (menu)		It is more applied on mobile application and quickly switch between top-level item sections of the view(Developer, 2023; Kalbach, 2007; Yang & Chiu, 2022).
Dropline Menus		When users hover their mouse over or click on the main menu item, a line will appear below the main menu in Dropline Menus, displaying additional options(Andrea, 2012).
Fat footer		It be found at the bottom of every page on the website nearly, assuming various forms that correspond to the website's content. Despite their diverse appearances, their importance remains paramount, albeit often overlooked. (and highly underrated)(Fessenden, 2019, 2020; Zheng, 2015). Giving a lot of navigation options at the bottom of the website, so this type of the menu is also called the “safety net”(Imetska, 2016).
Sitemap		This type of menu is the visualization tool that is used predominantly for planning purposes(Tankala, 2023). It offers users an alternative means of navigation, rescuing them from getting lost amidst the clutter of a crowded website(Group, 2008; Zheng, 2015).
Bread crumb		It is always a secondary navigation aid. Through one line of text to show a page's location in the website hierarchy(Kalbach, 2007; Laubheimer, 2018; Nielsen, 2007). Its modest nature that people are becoming accustomed to it(Nielsen, 2007).

According to a study, in the past, websites often used drop-down menus as the standard for navigation systems(Naylor, 2016). Therefore, in recent years, with the rapid development of the Internet, developers have begun to adopt different forms of menus in order to cater to the users' operating habits and enhance user experience. A collation of studies in 2017 found that more diverse forms of menus have appeared in recent years, no longer limited to the drop-down menus that were more monotonous and familiar in the past (Comeaux, 2017; Naylor, 2016). As menu formats show a great deal of

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diversity, they need to be further extensively researched and explored. In recent years, there has been a burgeoning interest in mega menus due to their capacity to present numerous options at a single level, exceeding the capabilities of traditional drop-down menus. While mega menus share some similarities with expansive hierarchical menus, two distinct differences set them apart: 1) mega menus can simultaneously display lower-level links within the same screen area while maintaining a comprehensive view of the menu's depth, enabling users to grasp the entirety of the menu structure clearly. 2) In a Mega Menu, the presentation of both breadth and depth allows users to directly navigate to their desired final goal (i.e., lower-level target) based on their specific needs. This direct access method facilitates users in reaching their final objectives efficiently without the need for sequential selections. The concept of menu level correlation was initially introduced by Miller in 1981. Experimental findings indicated that breadth surpassed depth in menu structure, resulting in the quickest search times and fewest errors when utilizing a two-level menu format (Miller, 1981). Subsequent to the replication of Miller's experiments by other scholars in 1983, the original study's limitations were addressed and rectified. This subsequent study also underscored that expansive menus yielded more precise and expedited search results (Snowberry et al., 1983). Furthermore, it has been demonstrated that broader menus structured in two levels yield the swiftest search times (Kiger, 1984).

Although mega menus had not been applied to websites prior to these studies (Naylor, 2016), they provided subsequent researchers with a conceptualization of the depth and breadth of menu levels. In 1988, scholars conducted a study involving six types of menus, which ultimately showed that optimal operational performance could be achieved through two levels of menus (Parkinson et al., 1988). And the prototype of the optimal menu has a similar layout to today's mega menus. In 2016, Naylor proposed that menu type does not affect search performance, a finding that is inconsistent with previous research (Naylor, 2016). However, taking an overview of the past studies, there are only preliminary discussions on mega menus, and most of the research experiments use the length of time to judge the efficiency, and fewer directly validate and discuss the efficiency of the menus in a more advanced way. Therefore, there are possibilities and extensions to be explored in this field. In this study, eye-tracking devices are used to sense the user experience and operation of the mega-menu.

2.2 User Experience (UX)

In 2006, Hassenzahl provided a comprehensive overview and analysis of user experience, illustrating that it extends beyond mere technique to embody a subjective, contextualized, intricate, and dynamic encounter (Hassenzahl & Tractinsky, 2006). Google defines UX as a phenomenon where empathy assumes a pivotal role, and it emphasizes that UX consists of usable, equitable, enjoyable, and useful attributes, all of which significantly impact both customers and businesses. Don Norman and Nielsen in 2016 provided a definition of user experience (UX) as the foremost criterion for fulfilling users' actual needs. This entails a comprehensive service design approach spanning various domains, including engineering, operations, graphics, industrial design, and interface design, etc. The study emphasizes the critical distinction between user experience (UX) and user interface (UI). While UI concentrates on the overall presentation of the interface and holds significant importance, UX encompasses broader aspects such as system accessibility, efficiency, and user enjoyment, etc. This differentiation underscores the necessity of separately addressing UX and UI for a holistic design approach. The usability of a website pertains to the procedural aspects that a user engages with while navigating a website. (Norman & Nielsen, 2016).

The primary objective of usability engineering is to comprehensively evaluate user interactions with a website and to enhance the human-machine interface(Nielsen, 1994a, 1994b, 2012; Nielsen & Levy, 1994; Nielsen & Pernice, 2010). Nielsen introduced fundamental concepts in usability engineering in 1994 and 2012, collaborating with Levy (1994) and Pernice (2010) on related research and developments. According to Webster and Ahuja (2006), the organizational structure

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of a website profoundly influences its efficacy and user sentiment (Webster & Ahuja, 2006). When users experience disorientation (Chang et al., 2022; Lazar et al., 2003; Levene, 2011; Nah & Davis, 2002). User Experience (UX) is akin to a comprehensive strategy aimed at safeguarding users while navigating a site, and mitigating potential harm they may inadvertently cause, recognizing that even minor elements can unsettle and disrupt their experience. Therefore, synthesizing previous literature, user experience encompasses an additional layer of operational efficiency compared to user interface. This study predicts that good operational efficiency will contribute to enhancing user experience, thereby increasing user satisfaction.

Nielsen advocated for observing user behavior by discerning the user's intentions or tasks, meticulously noting both failures and successes, and establishing conducive environments for users to articulate their opinions. Observation serves to comprehend the user perspective comprehensively, which is then complemented by substantive prototype testing and numerous prototype iterations to attain the ultimate design outcome. Minor issues frequently arise during operation, underscoring the importance of rigor in the process. Furthermore, attentiveness to user sentiments and thoughts is crucial. Consequently, this study employs user interviews to aid in elucidating the physiological sensory process. In 2022, a report by the N/N Group stated that eye-tracking devices are suitable tools for researchers to comprehend user behavior (Gordon & Rohrer, 2022). Given the progress and evolution of technology, coupled with the growing complexity of consumer behaviors (Rennie & Protheroe, 2020), eye-tracking devices have gained recognition as effective tools for investigating consumer viewing behaviors. Hence, it is opportune to incorporate eye-tracking devices in this study, aiming not only to yield more explicit data for exploring user experiences with menus but also to provide a visual reference for future researchers.

2.3 Eye-Tracking

The reason why research efforts have not burgeoned despite significant improvements in eye-tracking technology over the years is clearly explained in a study conducted in 2003. This limitation can be attributed to the necessity of individually calibrating eye-tracking instruments for each subject, coupled with the fact that approximately 10-20% of subjects may evade successful tracking. Moreover, a substantial amount of time is dedicated to the analysis and organization of data, with uncertainties persisting in data interpretation. For instance, discerning whether prolonged or frequent fixation on a subject by viewers signifies its importance or its complexity remains ambiguous (Jacob & Karn, 2003). In recent years, due to technological advancements, research in the field of eye tracking has proliferated. Recent studies indicate that during website browsing, people's visual attention typically conforms to an F-shape pattern (although some scholars have also proposed E- or Z-shape patterns) when scanning the entire website (Nielsen, 2006; Nielsen Norman Group, 2009; Shrestha & Owens, 2009). Moreover, scholars have posited that viewing priority typically begins at the top left corner, progresses to the top right corner, then moves to the bottom left corner, and finally to the bottom right corner. It has also been suggested that viewing behavior proceeds horizontally from left to right, followed by a vertical movement from top to bottom (Djamasbi et al., 2011; Moran, 2020). From an ocular perspective, the horizontal trajectory exhibits greater regularity compared to the vertical trajectory (Goldberg et al., 2002; Pan et al., 2004).

Namoun indicated in 2018 highlighted that when an object (task) is positioned within a fixed area, it heightens the concentration of attention during observation, consequently resulting in extensive coverage of the area of interest by eye-tracking, a phenomenon termed as depth scanning mode. The textual layout appears to influence the trajectory of the user's gaze during viewing, with a columnar layout being more aesthetically pleasing than a grid layout. Nonetheless, prior research has only tentatively suggested that a column layout garners visual attention, and its impact on information search efficiency remains inconclusive (Namoun, 2018). According to the "eye-mind" hypothesis proposed by Just and Carpenter,

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when an individual fixates on a particular passage of text or a specific region, the brain concurrently processes the corresponding information (Just & Carpenter, 1976). Put differently, eye movement data can furnish a dynamic trace of the locus of a person's attention within a visual display during observation (Hung & Wang, 2021).

In 2011, Leuthold, Schmutz, et al. employed eye-tracking devices to investigate disparities in users' visual perceptions among simple menus, dynamic menus, and vertical menus. The study's findings indicated that vertical menus require users to glance less frequently, operate more swiftly, and achieve a higher success rate. Consequently, this research posits that vertical menus are more conducive to exploring user perception and cognition (Leuthold et al., 2011). Nevertheless, the scope of eye-tracking applications in menu studies remains somewhat limited, underscoring the necessity for future studies to encompass a broader array of menu types. In 2003, Jacob and Karn highlighted the burgeoning interest in eye-tracking research within the domain of human-computer interaction, particularly its relevance to investigating user perceptions during menu item search processes. Their study synthesized findings from 20 usability studies, revealing a transition of focus from traditional computer menu navigation towards web search activities and information extraction from web pages. This evolution included a shift from targeted item searches within menus to exploratory browsing behaviors across websites without predefined objectives(Jacob & Karn, 2003). Jacob and Karn's insights underscore the utility of eye-tracking studies in informing navigation design and illuminating the role of eye gaze in information retrieval processes(Jacob & Karn, 2003; Nielsen & Pernice, 2010).

2.4 Phase Summary

Summarizing the foregoing literature review, this study aims to extend the future directions and address the gaps identified by previous scholars. First, it underscores the significance of the menu within the navigation system as a pivotal determinant of website usability. Second, it highlights the varying outcomes observed in prior research regarding the efficacy of expansive menus in facilitating user search processes, underscoring the need for more comprehensive data analysis in this regard. Thirdly, it advocates for a broader application of eye-tracking technology in menu research encompassing diverse menu types. Fourth, it advocates for the further expansion of eye-tracking utilization. Specifically, the integration of eye-tracking devices in menu research should encompass a wider spectrum of applications. Historically, empirical investigations have identified Eye-tracking Measurement Methods (EMMs) as effective tools for probing user search behavior and gaze patterns within navigation systems structured by information architecture (Nielsen & Pernice, 2010). Such methodologies enable profound analysis of physiological data, thereby enhancing researchers' comprehension of user behaviors. Furthermore, eye-tracking technology plays a pivotal role in experimental settings, facilitating post-experimentation recall through the playback of eye movements and supporting semi-structured interviews (Bartels, 2019).

This study commences by providing an initial clarification of the menu hierarchy, drawing upon Kalbach's (2007) conceptualization of the hierarchical organization of a website (Kalbach, 2007). Within this framework, the elements comprising the hierarchical structure are denoted as "Hierarchical Structure," with the uppermost tier designated as "Main Navigation," and subsequent tiers referred to as "Local Navigation." Thus, in alignment with Kalbach's hierarchical model, this study adopts the terminology of "Main Menu" for the primary level menu and "Local Menu" for the subordinate levels.

3 METHODOLOGY

This study underwent review and approval by the Ethics Committee at [National Cheng Kung University] (reference number: [111-306]). All subjects participating in the study provided informed consent and adhered to the pertinent ethical guidelines and safeguards.

This study integrates perspectives from developers and users, incorporating expert interviews to offer a comprehensive user-centric view. It evaluates the website’s strengths and weaknesses across various perspectives, aiming to synthesize insights to improve the website's operational experience and efficiency. Additionally, an eye-tracking

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experiment was conducted using an iterated version of the menu. Forty participants were invited to navigate the website in a laboratory setting, providing post-operation feedback on their psychological experiences for reference and data analysis purposes. The study is therefore divided into two phases:

Study 1: This phase involves investigating the current e-commerce menu format through user questionnaires and expert interviews, followed by facilitating interactions among users via focus groups. These focus groups aid in refining website redesign efforts by enhancing focus and elucidating operational challenges across diverse user backgrounds.

Study 2: This phase builds upon the findings of Study 1 to create an adjusted redesign version. Additionally, it will involve the participation of 40 test subjects in the experimental phase. Through discussions with Tobii experts on eye-tracking data, a more in-depth analysis will be conducted. This will lead to the proposal of more suitable versions for future developers to reference.

3.1 Study 1: Menu Iteration

This study collected samples from B2C retail e-commerce platforms in the top five e-commerce countries globally, namely China, the United States, the United Kingdom, and Japan, for questionnaire administration. The study utilized website listings from the United States and the United Kingdom among Western nations, and Japan and Taiwan among Eastern nations, for questionnaire purposes. E-commerce platforms spanning multiple countries were selected as representative examples for this questionnaire. Taiwan's e-commerce sector has exhibited a significant growth rate of 28.5% within a year (Min-jun, 2021), attributed in part to the preceding pandemic, indicating substantial potential for retail expansion. Consequently, forthcoming studies will incorporate iterative updates featuring Taiwan's e-commerce platform. The questionnaire was administered through the Maze (www.Maze.com), which is an online platform access users' usability on the website and was completed by the user. Simultaneously, the platform recorded error rates, mouse-click positions, and total time spent on each operation to offer researchers and experts an accurate benchmark for interviews.

3.1.1 Questionnaire Construction

This phase comprises four questionnaires, each assessing the usability and satisfaction of a cross-border e-commerce website. Simultaneously, we employed the backend of the Maze platform to collect data, encompassing metrics such as Homepage exit count (i.e., the number of users who abandoned the operation), Mis-click Rate, Avg Duration, Direct Success, Indirect Success as well as Give-up / Bounce. We leveraged this data to pinpoint areas where test subjects were prone to clicking incorrectly during operations, facilitating discussions with experts and the proposal of a prototype design for revision.

3.1.2 Expert Interviews

Four experts were invited to participate in this phase of the study. We invited one interface designer, two front-end engineers, and one back-end engineer, each with over three years of work experience. Initially, we commenced the interviews by gaining insights into their pertinent backgrounds and experiences. Subsequently, we delved into trend analyses and the perspectives of these experts across diverse fields concerning B2C cross-border e-commerce. Following this, we investigated the impact of information architecture within this industry. Ultimately, we engaged in discussions with the experts regarding the strengths and weaknesses of navigation systems across four e-commerce platforms. These discussions were informed by the results gleaned from user-administered questionnaires and an examination of the menu structures of each website.

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3.2 Study 2: Eye-tracking Experiments

3.2.1 Eye-tracking Experiments

Participant

Based on prior research, it is imperative to ensure a sample size exceeding 30 participants to establish a valid oculomotor thermogram (Nielsen & Pernice, 2010). Therefore, 40 participants were initially recruited for the within-subject design experiment conducted in this study. Subsequently, one participant withdrew from the experiment due to personal reasons, resulting in a final sample of 17 males and 22 females.

Material (Stimuli)

In this phase, we utilized Adobe XD for prototype design. We integrated feedback from previous experts and users across four B2C e-commerce platforms to inform our prototype designs. The experiment employed purposive search techniques to effectively manage task time, during which we focused on identifying a vacuum cleaner robot as the target purchase. To avoid any bias stemming from participants' familiarity with robot vacuum cleaners, we excluded all product logos from the images used in the study. Additionally, we selected robot vacuum cleaners with similar appearances to minimize the potential for brand recognition by the participants. Each participant interacted with two versions of the navigation system, one featuring the initial horizontal design iteration and another featuring the subsequent horizontal design iteration, as well as the initial horizontal design iteration paired with the subsequent vertical design iteration. Before commencing the experiment, we delineated the Areas of Interest (AOI) on both versions, streamlining subsequent data analysis, particularly in evaluating menu efficiency. The following is a schematic diagram of the AOI (Figure 1):

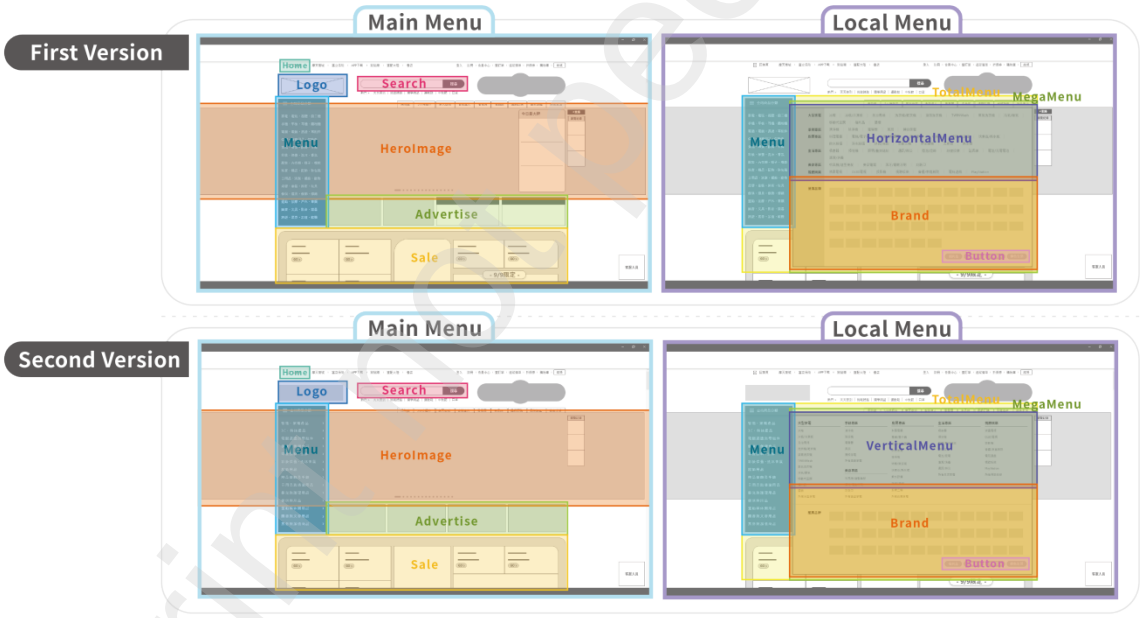


Figure 1 Menu Definition and Prototype

Measurement

In their 2003 study, Jacob and Karn noted that the definition of eye-tracking data lacked clarity. However, recent years have witnessed rapid advancements and maturation in eye-tracking technology, leading to its widespread utilization in

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research (Holmqvist et al., 2011; Klein & Ettinger, 2019; Nakano et al., 2013; Nielsen & Pernice, 2010; Thoma & Dodd, 2019; Van Gompel, 2007). Eye-tracking devices have proven effective in aiding researchers' understanding of eye movement patterns during reading (Van Gompel, 2007). Moreover, the neural circuits linked with rapid eye movements have undergone extensive study in the medical field, particularly in humans where they are well-documented, while similar physiological investigations have been conducted in non-human primates (Van Gompel, 2007).

Previous studies have compiled and analyzed the applications of eye-tracking in various fields (Nielsen & Pernice, 2010; Van Gompel, 2007). Concerning human-computer interaction, research related to eye-tracking is also progressively emerging. In a 2013 publication (p. 117) co-authored by Nakano, Conati, and Bader, it is noted that there are nearly fifty extractable eye-movement features in eye-tracking device data. However, there remains a question as to whether a more concise set of features can be employed to achieve comparable performance and thereby enhance computational efficiency (Nakano et al., 2013). Hence, this study utilized data from Tobii Pro Lab to analyze five aspects, including Visit Duration, Fixation Duration, Total Time of Interest Duration, Revisit Count, and Reach People. Additionally, the study engaged in discussions with experts from Tobii to devise various methods for in-depth data exploration. The data of this study is defined as Table 2 below:

Table 2 The metric definitions of this study

Metric	Definition
Heat Map	It shows the combined fixations of total participants on the pages(Nielsen & Pernice, 2010).
Gaze	One gaze plot shows a single participant's visit on the page. Each plot dot means a fixation. More larger dot mean more longer fixation. The lines between the plot dots are saccades(Nielsen & Pernice, 2010; Tobii, 2023).
Fixations	During fixations, the eyes come to a halt, ceasing the scanning of the visual scene, and remain relatively immobile. This allows for maintaining focus on a stationary object of interest directly on the fovea, facilitating detailed intake of visual information(Nielsen & Pernice, 2010; Tobii, 2023).
Attention Ratio	It is defined as a measure of the proportion or degree of focus allocation by individuals or groups in a specific context or task. Fixation Duration/Total Time of Interest Duration
Effective Focus	It can be defined as the ability of individuals or group to concentrate and utilize time and resources efficiently while accomplishing tasks. Fixation Duration/Visit Duration
Reach Rate	It refers to the percentage of the target participants reached or engaged within a specific timeframe. Number of People Paying Attention to this AOI / Total Number of Participants
Revisit Rate	It refers to the percentage of participants who revisit a AOI within a defined period of time. Revisit Count / Reach People

Procedure

Before the commencement of the experiment, researchers will prepare the task of locating the target for this study. This research adopts a purposive search strategy based on the experiences of previous scholars to allow researchers to control the operation time and to prevent excessive eye use by the participants, which could lead to eye fatigue. According to Tobii's technical experts, when conducting eye-tracking experiments, it is advisable to limit the duration to within 30 minutes to avoid potential data quality degradation due to eye fatigue in the later stages. Prior to the start of the experiment, researchers will describe the shopping scenario to the participants: "*Today, the robot vacuum cleaner in your home is broken, and you plan to purchase a new home appliance through online shopping. You have decided on an iRobot model robot vacuum cleaner and will search for the product using the website menu. This experiment does not have a time constraint; please proceed in the most comfortable and relaxed posture*."

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In 2019, Tobii published a web article indicating that conducting thorough user interviews while utilizing an eye-tracking device facilitates post-analysis of collected data. Nonetheless, given that thinking aloud can evoke a sense of artificiality in subjects, we advocated for a gentle encouragement of thinking aloud without coercion, thereby permitting subjects to supplement their insights during task manipulation. This methodology, complemented by video playback at the experiment's conclusion, rendered the detailed retrospective discussion more organic for both subjects and researchers alike. Before commencing the experiment, every participant received detailed instructions from the researcher to complete an informed consent form, ensuring their comprehension of the procedures and rights involved. At the onset of the experiment, the purchase scenario appeared on-screen, accompanied by the researcher's verbal description of the task, ensuring its clear and effective communication to the participant. The researcher maintained non-intervention during task execution, and participants subsequently completed usability scales and satisfaction questionnaires after each task. Lastly, eye-movement playback was employed to facilitate Retrospective think aloud methodology, RTA (Retrospective think aloud methodology, RTA) of cognitive processes. According to Tobii's proposal, participants involved in situational tasks often exhibit more manipulative behaviors. Consequently, employing the Concurrent Think Aloud methodology (CTA) may result in an unnatural level of attention. Ideally, the Retrospective Think Aloud methodology (RTA) should be utilized for qualitative data collection. This approach enables the participant to concentrate on the eye movement experiment before engaging in the RTA(Bartels, 2019). The researcher can playback a video of the gaze trajectory during manipulation, enabling the participant to articulate their thoughts during the manipulation task while observing the video. Additionally, the brief RTA session facilitates a suitable transition and adaptation period for the participant between the two tasks. Figure 2 shows the stages of the experiment.:

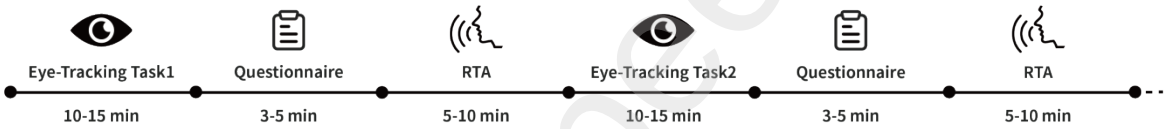


Figure 2 The experimental procedure

4 RESULTS

4.1 Study 1: Iteration of Menu Prototypes

4.1.1 Results: Users And Experts

A total of 101 questionnaires were collected in this phase, with 100 valid questionnaires. Through the menu manipulation of these questionnaires, we initially explored the data provided by the backend of the Maze platform (see Table 3). In the UK, users spent less time on menu manipulation and had a higher success rate (78.3%). In comparison, the US had a lower direct success rate, but it is worth noting that it excelled in indirect success rate, which was the best among the four sites. This implies that the website possesses the capability to assist users in retracing their steps to the correct path upon becoming disoriented. In the Taiwan segment of our study, we observed faster operation times and higher success rates. However, we hypothesized that these outcomes might be attributed to respondent ethnicity, with a majority being familiar with the Taiwan website. Analysis of open-ended questionnaire responses revealed that despite the higher success rate on the Taiwan website, respondents generally provided less favorable feedback. Common complaints included the complexity of menu navigation, overwhelming information volume, and unclear categorization. User feedback predominantly centered on challenges related to menu complexity and overall site usability.

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Table 3 Data about menu manipulation

Country	Homepage exit count	Misclick Rate	Avg Duration	Direct Success	Indirect Success	Give-up / Bounce
United State	3	45.7%	14.1s	43.5%	**21.7%**	34.8%
United Kingdom	4	37.8%	16.4s	**78.3%**	0%	21.7%
Japan	8	48.3%	27.3s	67.7%	0%	32.3%
Taiwan	4	28.3%	16.6s	73.9%	0%	26.1%

The primary objective of information architecture is user-centricity. It is often stated by experts that while a good User Interface (UI) is crucial, it does not automatically translate to a good User Experience (UX). However, UI serves as the initial point to captivate users, with UX following to create a lasting positive impression. Only when both aspects are well-executed can we effectively retain customers. In this research endeavor, we have focused on restructuring information through text layout optimization and revamping the original menu specifically for Taiwan's version. Drawing from previous eye-tracking studies, we identified the F-shaped reading pattern as a significant factor in iterative design considerations for webpages. Our hypothesis posits that implementing the F-shaped visual hierarchy onto webpages, particularly in navigating from the main menu to submenu areas, can enhance operational efficiency and usability, thus improving overall user experience. At last, we have crafted two prototypes tailored for a Known-item seeking task, intended for the subsequent phase of the eye-tracking experiment. In the original version (version 1), both the main menu and the local menu were arranged horizontally. In response to feedback from users and experts, the second version retained the horizontal alignment for the main menu, albeit refined into more succinct categories. Notably, the local menu was reconfigured to a vertical orientation. Figure 3 shows a difference between the two versions of the menus:

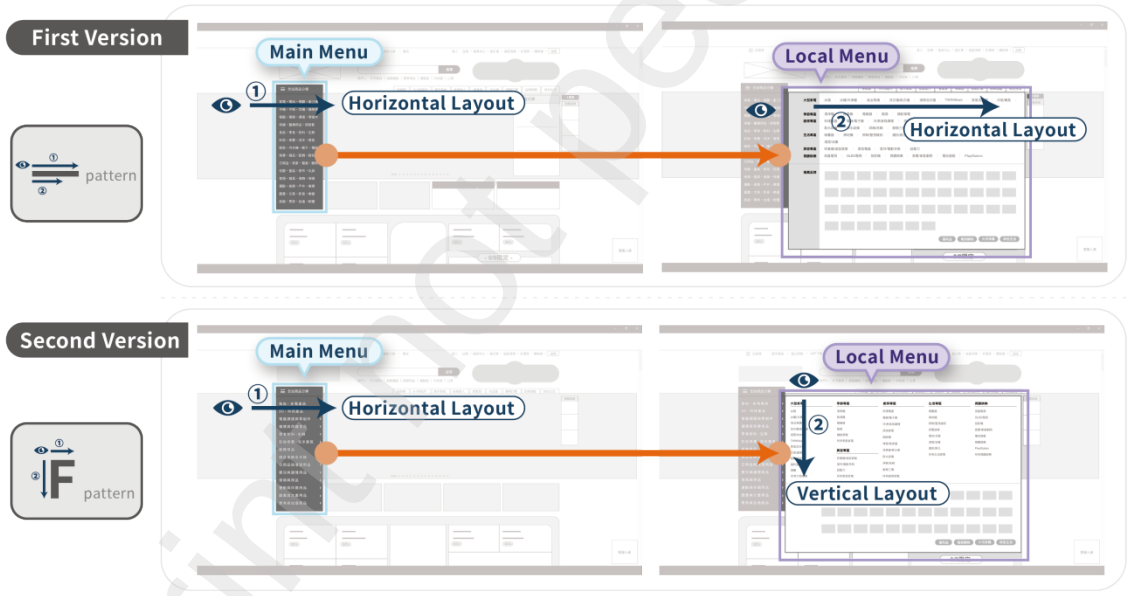


Figure 3 Two Versions Menu Layout

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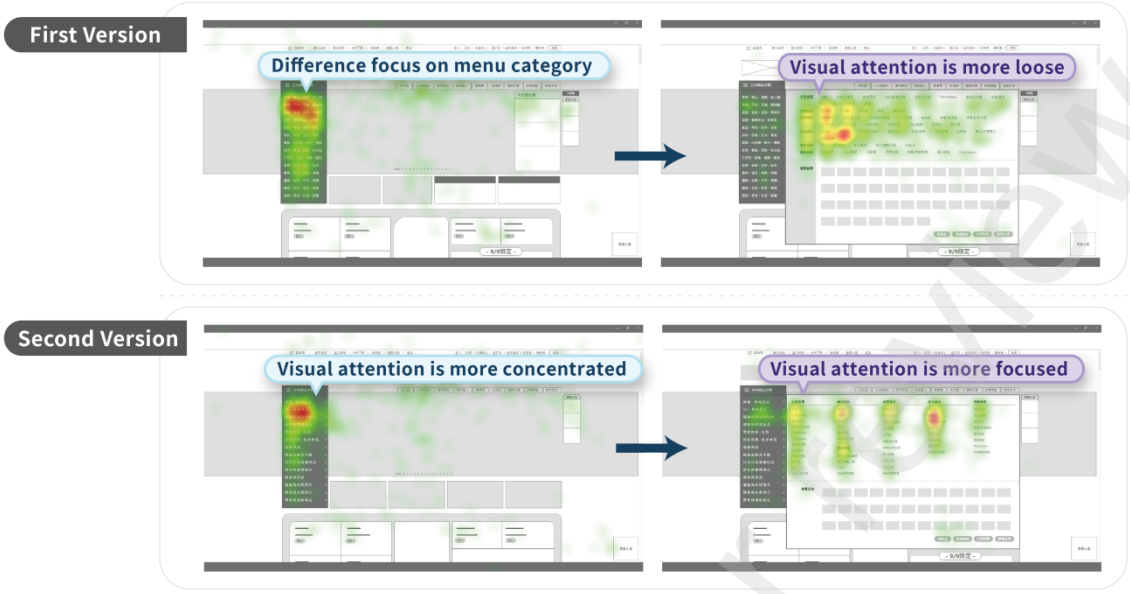


Figure 4 Two Versions of Heat Map

4.2 Results of Study 2

4.2.1 Usability and Satisfaction

Satisfaction and usability experienced significant improvement between the two versions iterations of the menu. Previous studies have established a positive correlation between usability and satisfaction. Moving forward, we aim to delve deeper into the connection between the main menu and local menu concerning their impact on satisfaction and usability flow. Table 4 shows the correlation coefficients between initial usability and satisfaction between the two versions.

Table 4 Pearson correlation coefficient Matrix between SUS and Satisfaction.

Version		SUS	Satisfaction
First Version	SUS	1	0.762**
	Satisfaction		1
Second Version	SUS	1	0.750**
	Satisfaction		1

** Correlation is significant at the 0.01 level (2-tailed).

* Correlation is significant at the 0.05 level (2-tailed).

4.2.2 Heat Map

The preliminary heat map results indicate that red areas in the heat map of the main menu correspond to points of interest for multiple subjects. In contrast to Version 2, the main menu in Version 1 was less populated. Initial interviews suggest that the menu categories in Version 1 might have been overly complex, leading to confusion among respondents during navigation. A more thorough data analysis would be carried out in the subsequent phase. In the open local menu, the horizontal layout of the initial version increased difficulty in readability and caused subjects to become disoriented

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when searching for specific subcategories. Past research suggests that such disorientation can lead to confusion and frustration among subjects (Chang et al., 2022; Lazar et al., 2003; Levene, 2011; Nah & Davis, 2002). The preliminary heat map also revealed that changing the menu to a vertical format (Version 2) allowed subjects to focus better on larger blocks of information. Therefore, we tentatively propose that a vertical menu format may enhance subjects' ability to navigate information effectively. Figure 4 illustrates these findings.

4.2.3 Data Analysis: Main Menu

In the data analysis pertaining to the main menu, distinctions in Fixation Duration and Effective Focus were initially assessed through an independent samples t-test. In the descriptive statistics, each version had 39 participants. In terms of fixation duration, a significant difference was observed between Version 1 ($M = 8.75$, $SD = 9.49$) and Version 2 ($M = 4.82$, $SD = 3.58$) (as in Table 5). Based on the results of the RTA, the subjects noted that the menu categories in the initial version were overly subdivided. This resulted in numerous errors as they struggled to locate the appropriate category during the task. Hence, we proceeded to deliberate on the efficacy of the simple category (Version 2) within the main menu, aiming to augment users' operational experiences while elucidating notable variances in Effective Focus. Table 6 illustrates significant discrepancies in Effective Focus between Version 1 ($M = 0.90$, $SD = 0.18$) and Version 2 ($M = 0.36$, $SD = 0.39$). The revision in Version 2 significantly improved overall viewing efficiency, as evidenced by a higher performance in attention span and viewing return rates compared to the condensed category (Version 2) (refer to). Interviews with subjects revealed that excessive description and categorization heightened attention and Revisit. Their frequent revisits were primarily due to uncertainties in the operational process. Subjects' repeated visits to the main menu not only demonstrated hesitancy but also suggested that a higher proportion of returns to the main menu might lower viewing efficiency.

Table 5 Independent Samples T-test of the two version of Fixation Duration and Effective Focus.

Version	t	df	p	Mean Difference	SE Difference
Fixation Duration	2.421	76	0.018	3.930	1.624
Effective Focus	8.078	76	0.000	0.540	0.067

Specifically, as observed in Figure 5Effective Focus, the first version shows higher values compared to the second version. According to the indicator definitions outlined in this study, Effective Focus is calculated based on the attention duration and viewing time within the AOI (Area of Interest) drawn region. In comparison to Figure 5, the eye-tracking data for the first version predominantly covers the entire AOI of the main menu. Conversely, in the second version, users' eye-tracking paths are more intuitive and focused within the AOI due to the clear description of menu categories (resulting in a more concentrated heat map). Regarding Table 7, we hypothesize that the elongated Attention Ratio in the initial step of the first version may contribute to a certain level of user satisfaction with the website.

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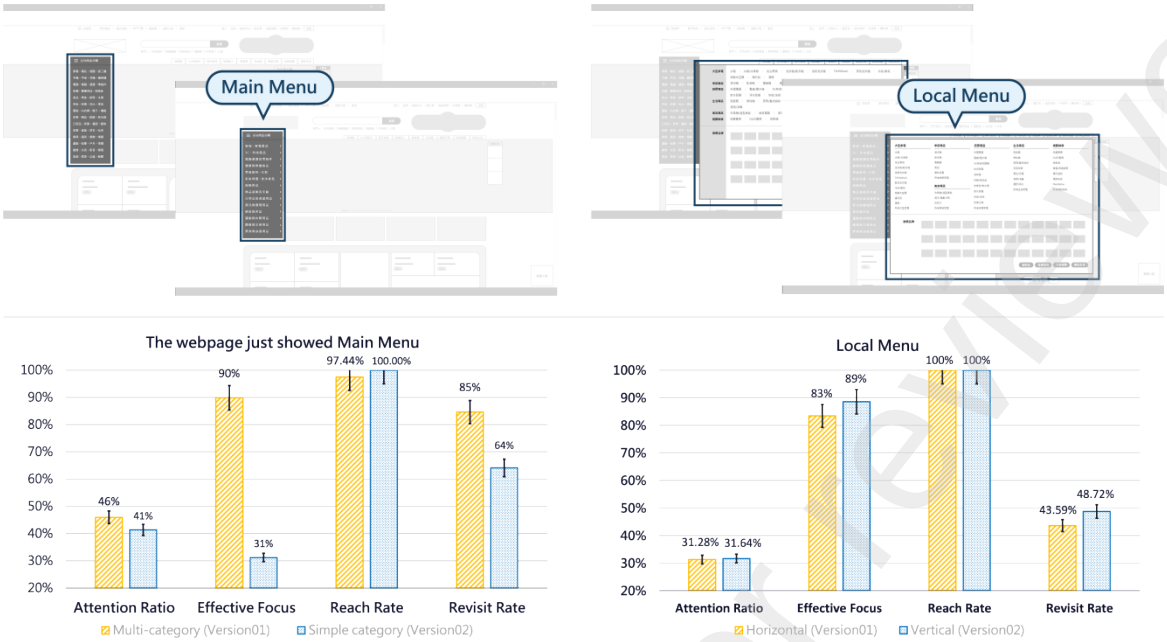


Figure 5 Data of Main Menu and Local Menu

4.2.4 Data Analysis: Local Menu

Regarding the local menu, we found that only the local menu in version 1 had no significant differences in usability and satisfaction with the physiological perceptions of the participants when they entered the local menu (i.e., clicked on the local menu). Notably, in the Second version, the local menu (Total Time of Interest Duration, TTID) was presented in a vertical orientation, leading to a negative correlation with satisfaction and usability (refer to Table 6), yet resulting in a positive correlation with the time spent in the main menu. The current study posits a hypothesis in line with prior expert theories: that User Interface (UI) serves to attract users, while User Experience (UX) encourages user retention. Engaging users at the main menu tends to prolong their subsequent stay on the following page local menu. Nonetheless, prolonged stays on the local menu can potentially result in diminished SUS and overall satisfaction.

Based on the initial data analysis, notable correlations were identified within the redesigned interface. Subsequent data analysis revealed that without activation of the local menu (i.e., Table 7 Main menu), Attention Ratio exhibited a positive correlation with user satisfaction. Conversely, upon activation of the local menu (i.e., Table 7 Local menu), the Effective Focus demonstrated a positive correlation with usability. Observations from users' Retrospective Think Aloud (RTA) methodology indicated that when accessing the local menu, the vertical menu proved more effective for navigation and locating target information, resulting in reduced time spent on the main menu. Consequently, the higher revisit rate in this context (e.g., Figure 5) implies that vertical menus facilitate navigation and information retrieval—a departure from the wandering revisit rates observed with only the main menu present (e.g., Figure 5). Furthermore, independent sample t-tests were conducted to validate the hypothesis and assess differences between the two versions regarding regional menus in this study. The findings revealed a significant disparity between the first version (M = 0.84, SD = 0.15) and the second version (M = 0.89, SD = 0.05), t (76) = -2.082, p = .041, confirming the presence of distinct characteristics in regional menus across the versions.

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Table 6 Pearson correlation coefficient Matrix for Version Two.

Version	Main Menu- TTID	Local Menu-TTID	SUS	Satisfaction
Main Menu- TTID	1	0.559**	-0.122	-0.157
Local Menu- TTID		1	-0.362*	-0.349*
SUS			1	0.750**
Satisfaction				1

Total Time of Interest Duration, TTID

** Correlation is significant at the 0.01 level (2-tailed).

* Correlation is significant at the 0.05 level (2-tailed).

Table 7 Pearson correlation coefficient Matrix between Main Menu and Local Menu.

Page		Attention Ratio	SUS	Satisfaction
Main menu[a]	Attention Ratio	1	0.246	0.364*
	SUS		1	0.750**
	Satisfaction			1
		Effective Focus	SUS	Satisfaction
Local menu[b]	Effective Focus	1	0.322*	0.226
	SUS		1	0.750**
	Satisfaction			1

[a]Main menu: The webpage just showed Main Menu.

[b]Local menu: The webpage simultaneously showed Main Menu and Local Menu.

** Correlation is significant at the 0.01 level (2-tailed).

* Correlation is significant at the 0.05 level (2-tailed).

5 FINDINGS SUMMARY

After adjusting to the feedback from users and experts, the update has been effective in improving the efficiency of users in ordering meals. In Study 1, we found that user feedback was more concerned with the complexity of the menu categories and the overall experience. After discussing with experts, we constructed an idea that "UI is the first step to attract users, and UX is the second step to make a good impression. With both UI and UX strategies, we can drive customers back." Therefore, through user feedback and previous eye-tracking results, we found that users are good at viewing flat surfaces in an F-shape. Combining visual feedback and previous objective eye movement behaviour results, we ultimately recommended that the main menu and local menu to be presented in a combination of horizontal and vertical formats in the Study One. The results of Study One provided a good basis for a follow-up eye-tracking study, and through feedback from experts and users, we iterated on a second version of the (horizontally-integrated-vertical) design.

In Study 2, we divided the discussion into two parts: the main menu and the local menu. For the main menu, the result showed that too much detail in the menu would increase the time required for users to understand and guess, which may cause uncertainty in users' search for information. In the second version of the main menu, we simplified the layout of the menu that helped improve users' comprehension and viewing efficiency, according to the study. As for the local menu, it was found that the original version of the menu did not have a significant effect on usability and satisfaction; however, in the updated version of the menu, the local menu improved the efficiency (Effective Focus) of the users in finding information.

The findings suggest that employing straightforward and succinct menu designs for main menus effectively enhances users' attention. Increased attention on main menus correlates with a higher likelihood of subsequent exploration, including

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engagement with local menu. Moreover, the research revealed that heightened interest in main menus corresponds with greater user satisfaction, subsequently prompting engagement with local menu. However, it is important to note that if the local menu takes too long to unfold, it may diminish user usability and satisfaction with the website. Lastly, this study introduces a novel perspective for understanding the user's dynamic operational process within a navigation system. By employing the previously utilized flat F-type heat map along the user experience viewing line, dynamic visual breakpoints are identified through the amalgamation of horizontal and vertical directions, aimed at amplifying user operational efficiency and experience.

6 CONCLUSION AND DISCUSSION

In this study, mega menus are defined as those that can display lower-level links within the same screen area, providing a clear view of the entire menu structure. This presentation of both breadth and depth enables users to efficiently navigate directly to their desired targets, bypassing sequential selections. Drawing upon previous scholarly concepts of hierarchy (Kalbach, 2007), this research explores both main menus and local menus.

Therefore, it is important to recognize that excessive reliance on local menu may result in diminished user usability and satisfaction with a website. This study introduces a novel perspective on user dynamics within a navigation system. This approach adapts the traditional flat F-shaped heat map model to the user experience viewing line and establishes visual breakpoints through horizontal and vertical integration, thereby enhancing operational efficiency within the user experience. Menus represent a prevalent method for website navigation, comprising a series of links to various pages that collectively form the website's information structure, significantly influencing its usability (Naylor, 2016). Leuthold et al. (2011) note that vertical menus lead to reduced gaze frequency, faster navigation speeds, and higher success rates, underscoring their suitability for perception and cognition(Leuthold et al., 2011). Future research endeavors should encompass diverse navigation systems to tackle the multifaceted challenges confronting present and future users in navigation tasks. When confronted with intricate tasks, users tend to adapt navigation strategies (Leuthold et al., 2011) and tailor their behavior according to their own experience or the information acquired during information retrieval (Jul & Furnas, 1997; Zheng, 2015). Fenichel (1981) was one of the earliest to indicate that users' website search experience influences their search performance (Fenichel, 1981; Kim, 2001). Hence, building upon previous research, this study centers on enhancing user experience efficiency through menu design to alleviate users' cognitive processing load during information retrieval. Previous studies have frequently discussed the phenomenon of individuals following F or Z-shaped trajectories during visual perception, primarily within static planar viewing contexts. This study extends the concept of F-shaped trajectories to dynamic operations through experimental manipulation. In the design of the main menu, a horizontal layout is more effective in capturing viewers' attention, while for the local menu, a vertical F-shaped layout is integrated into the design considerations. As anticipated, the second version of the local menu presented in a vertical format demonstrated superior visual efficiency. This outcome reinforces the critical role of UI design in initial user engagement, with UX being paramount in user retention.

This study synthesizes input from both users and experts, providing a foundational framework for subsequent research on the validation of Version 2. Initial heat map analysis clearly reveals that participants exhibited enhanced eye focus when presented with a vertically oriented local menu. Additionally, our investigation found a marked decrease in concentration at the first level of the main menu, in contrast to the engagement elicited by static images. Considering the concept of 'Known-item seeking,' users typically seek information in the most efficient and expedient manner possible to facilitate decision-making. We aim to reduce the time spent in the search process overall. Version 2 of the platform demonstrated enhanced efficiency in viewing and proved to be more effective than the multi-category format, as indicated by statistical

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analysis of viewing time and return viewing percentages. Retrospective interviews revealed that the multi-category menu caused confusion among respondents, leading to increased navigation time and feelings of confusion. Concerning local menu, users tend to access them more frequently when they find the content of the main menu intriguing. Nevertheless, when users navigate these local menus to seek information, the efficiency of the menu interface plays a crucial role. Investing an appropriate amount of time in navigating local menu can enhance usability and user satisfaction. Conversely, prolonged search times may diminish the user's perception of the site's usability and satisfaction levels. Skillful design of local menu further enhances user efficiency, subsequently contributing to the overall smoothness of site navigation (refer to Table 7).

Based on prior research, this study investigates the contributions of mega-menus to navigation systems within information architecture, delineated as follows: (1) a comprehensive assessment of the efficacy of mega-menus in enhancing users' search efficiency, utilizing eye-tracking data; (2) a more thorough and multifaceted exploration within the realm of menu design; (3) precise delineations of each menu type; and (4) the integration of the visual trajectory (F-shaped) of the former flat plane into the dynamic operation process to present the main and local menus both horizontally and vertically. Furthermore, the study underscores the significance of usability within navigation systems (Palmer, 2002; Zheng, 2015). Despite the technological boom in the field of human-computer interaction, comprehensive research on menus remains scarce. While most users rely on search systems to locate information, menus serve as navigational aids akin to maps. In situations where search systems are unavailable, navigation systems become crucial tools for information retrieval. Despite being a passive tool, navigation systems are indispensable, enabling users to navigate websites with confidence. Even in the event of search system failure, navigation systems offer a dependable alternative.

"Is Beautiful Design Correlated with Enhanced Usability?" (Tractinsky et al., 2000). While it is acknowledged that an aesthetically pleasing interface can evoke positive user sentiments, it constitutes merely a fraction of the overall user experience. The functionality of the interface gains prominence once users engage with it. Consequently, as asserted by experts, "User Interface (UI) serves as the initial enticement for users, while User Experience (UX) is pivotal in fostering a favorable impression. It is only when both elements are seamlessly integrated that customer retention can be effectively achieved." This study contributes valuable insights for subsequent developers based on feedback from both experts and users. A positive user experience plays a crucial role in shaping a favorable perception of a brand (Krug, 2000; Naylor, 2016; Rosenfeld & Morville, 2002). Given the multitude of elements within a website that can influence user behavior, it is imperative to meticulously consider the impact of each detail during the development phase. This consideration becomes even more pertinent in light of the proliferation of cross-border e-commerce enterprises, particularly influenced by recent epidemics. Consequently, a comprehensive focus on the overall user experience becomes imperative not only for enhancing business gains but also for fostering customer loyalty retention.

This study focuses on exploring the menus of B2C retail e-commerce platforms. Given that the retail e-commerce industry encompasses a diverse and extensive range of products and services, it is more suitable to present the overall menu using mega menus to increase user operational efficiency. Consequently, this study only investigates the current trend of mega menus, lacking a broader exploration. However, due to the rapid development of internet technology, many different types of menus have been phased out without in-depth exploration. While traditional menus have disappeared in the fast-evolving internet era, they may provide valuable insights for future menu architecture. Although this study conducts a literature review and defines traditional menus, it has not performed further user research on various menu types. Future research should focus on reviewing relevant studies and comprehensively exploring different types of menus. Additionally, to simplify the investigation of how menu formats are perceived and experienced psychologically by users, this study overlooks related website elements such as color schemes, advertising displays, and brand influences. However,

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constructing a complete e-commerce platform relies on the integration of various elements. Therefore, future in-depth studies on menus should timely incorporate related elements to understand the mutual influences between menus and other elements. Lastly, catalyzed by the COVID-19 pandemic, cross-border e-commerce platforms have become battlegrounds among businesses. Under different cultural backgrounds, the development and planning of cross-border e-commerce may also emerge as an indispensable element. In B2C e-commerce platforms, the retail industry is the most complex and information-rich. Therefore, this study focuses more on exploring mega menus in B2C retail e-commerce platforms. However, it lacks a comprehensive discussion of other types of websites, such as brand websites and showcase websites.

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